

DT8063 – Lab Introduction

Canberk Ozen 4 September 2026

Based on the slides of Alexander Galozy, 4 September 2025

Overview – What and How

What?

- 4 Labs total
 - 1 lab scientific writing, 3 programming labs
- One session per lab, one session every week.
 - This is for your benefit to ask questions and work on the lab.
 - Take advantage of it – Tips and tricks you will receive here!

How?

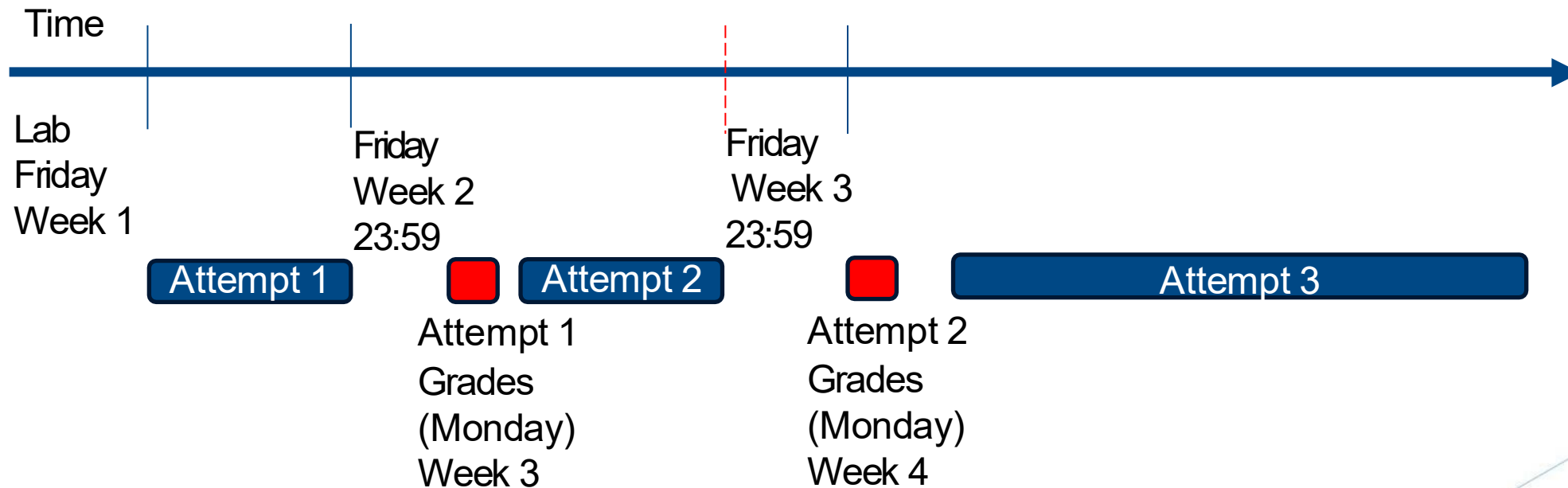
- **All of the four labs must be completed individually, so no group work is allowed.** So, if any two students submit the same notebook, it will be considered cheating. Be mindful about this to prevent any unpleasant conversations with the disciplinary board.
- NOTE: In general, for all labs, the use of Large Language Models (ChatGPT, Gemini, etc.) or any automated software to solve the problems outright is *prohibited* – You are here to learn. It is fine to use it if you need to correct language or get inspiration for your tasks. – **If suspected that such tools were used to avoid work outright, I reserve the possibility of oral examination of labs.**



Overview - Grading

- Grading?
 - 3 attempts (graded) per lab.
 - All attempts are to be done within the course timeframe – Exceptions only with good reason.
 - First attempt: The due date for the first attempt is always on Friday 23:59 the following week of the lab.
 - Graded next Monday.
 - Second attempt: Due date on Friday 23:59 the following week of the lab.
 - Graded next Monday.
 - Third attempt: End of the Course.
 - Late submissions will remove one attempt. I urge you to take advantage of submitting even if your lab is incomplete. You may still get useful feedback so your second/third attempt can benefit!

Overview - Grading



REMEMBER: Late submissions will remove one attempt!

Lab 1 – Scientific Writing



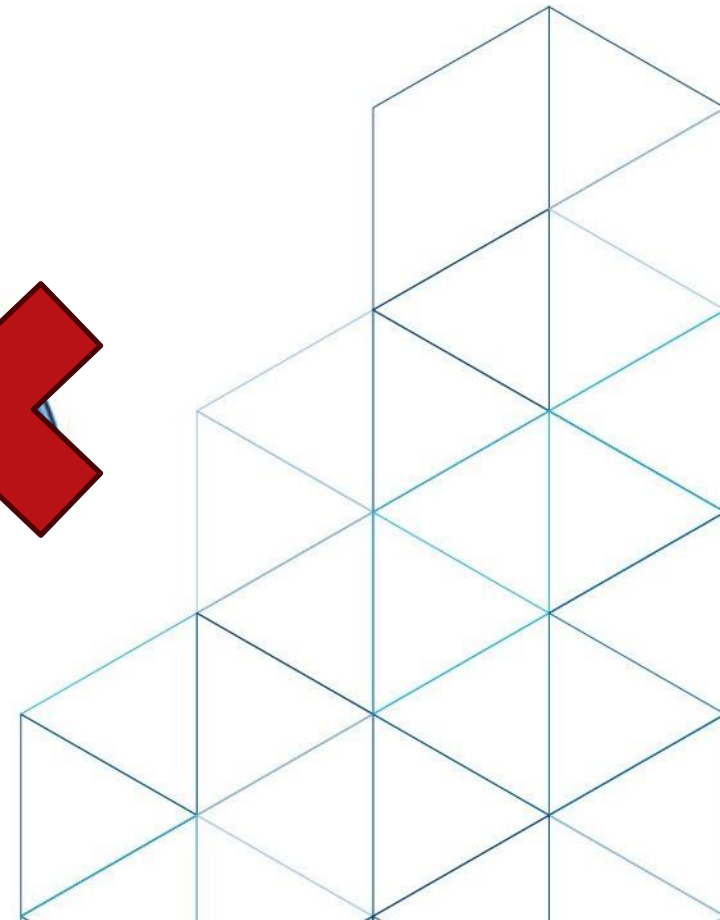
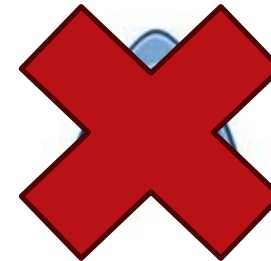
Which one of you use primarily a text-processor like Word/LibreOffice/OpenOffice etc. for your assignments?



Lab 1 – Scientific Writing



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Lab 1: Scientific Writing

- In this course we will use $\text{L}^{\text{A}}\text{T}_{\text{E}}\text{X}$
- Primary textprocessor in scientific writing
- In the first lab you will recreate a PDF document using LaTeX
 - Important! Writing in LaTeX is a little bit like programming. You will need to use specific commands to insert figures, tables and equations. Also, there is a specific way of referencing these in the text via – you guessed it – specific commands. *No manual labeling!*
- Overleaf handles all the compilation for you and results in easier writing sessions. Create an account in [overleaf](#) before starting the Lab 1.

Lab 1: Scientific Writing

- A bit of self-study required. You will need to read part of an online course on principles of scientific writing:
 - <https://cgi.duke.edu/web/sciwriting/index.php>
- Subsequently, there will be sentences provided that need to be changed to conform to the principles of scientific writing.
- Just a Reminder: This lab is a primary *target* by students for LLMs. The use of such in this lab is completely prohibited.
- You are allowed to make mistakes here. You will not fail the lab because some (a resonable amount) sentences are wrong.

Labs 2 – Basic Data Science with Python

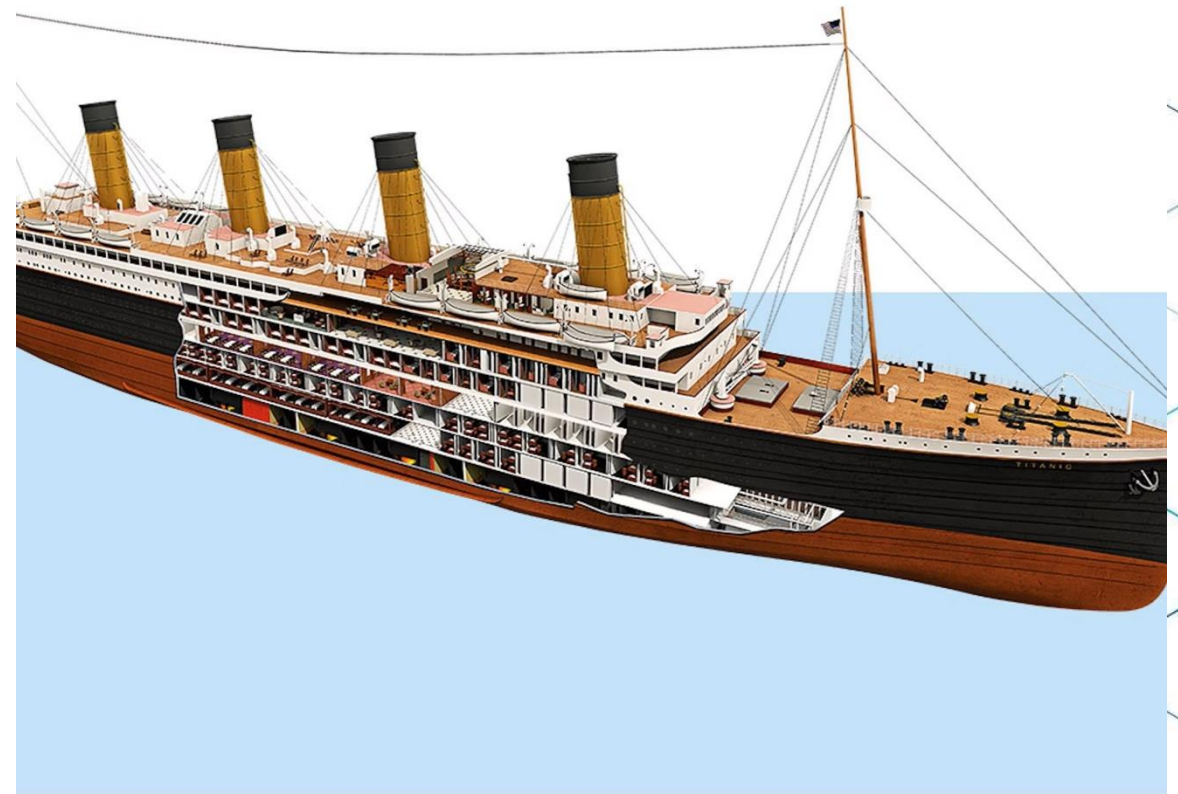
- For the programming labs we recommend you to use python with anaconda:

<https://www.anaconda.com/download>

- All labs use Jupyter notebooks!

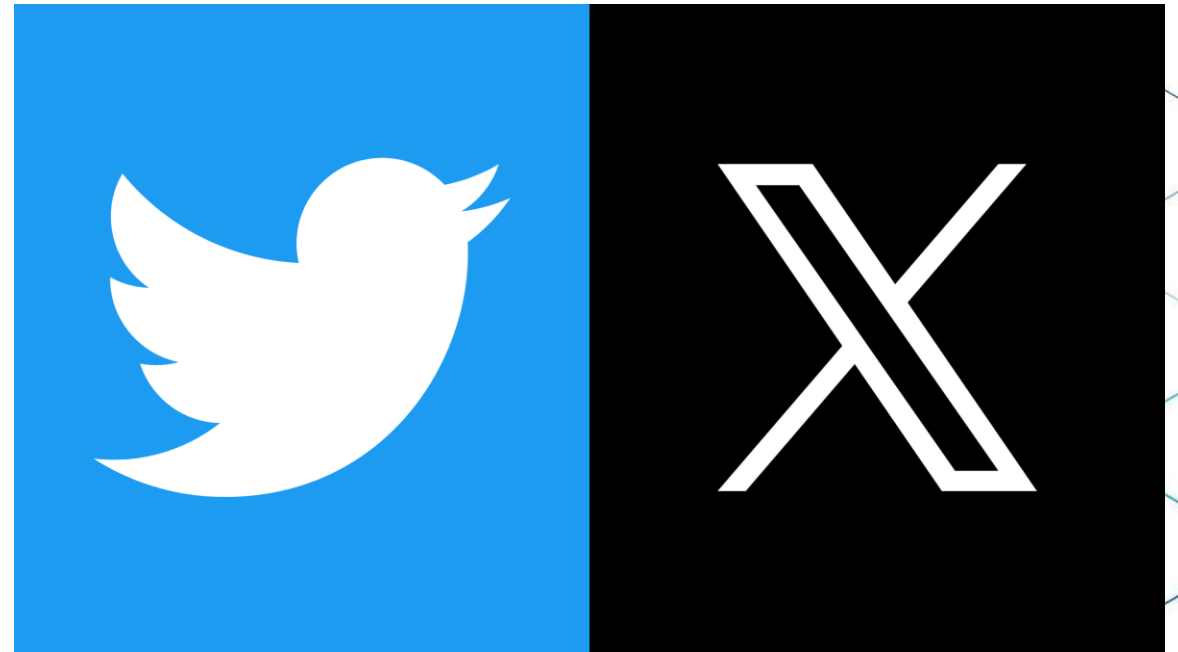


- This lab is about the data science pipeline, like data cleaning and imputation, as well as exploring the Titanic dataset and extracting some interesting facts about the disaster.



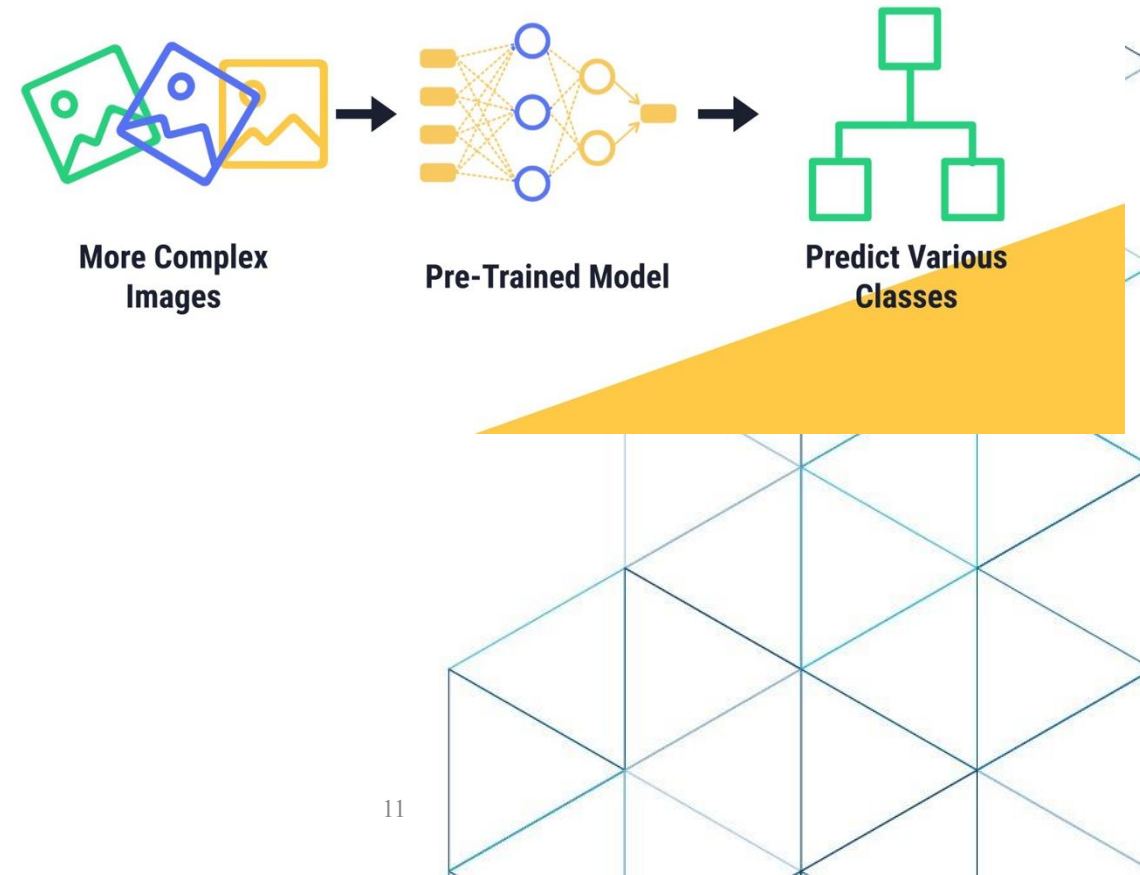
Labs 3 – Text Processing

- In this lab, you will do some text processing and analyze twitter/X tweets!
- Pre-processing tweets and gaining some interesting insights from user interactions.



Labs 4 – Neural Networks off-the-shelf and finetuning

- In this assignment, you will work with pre-trained Neural Networks that can be used for prediction tasks via transfer learning and finetuning on both text and images.
- We will use Google Colab for this lab.
- <https://colab.research.google.com/> **colab**
- You don't need to use this! It just makes it easier. You are free to work on it using your own computer.



Questions?
Suggestions?

